GURU NANAK DEV ENGINEERING COLLEGE, LUDHIANA

Department of CSE

ASSIGNMENT PLAN

Name of the Course-Coordinator: **Dr. Kapil Sharma, Dr. Satinderpal Singh**
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Branch	Semester	Subject	Code	Session
CSE	3rd	Computer Architecture	CCS203	Aug – Dec, 2025

ASSIGNMENT NO. 1

Course-Content Covered:

Unit-1 Data Representation: Data types, Complements, Fixed point representation, IEEE 754 Floating point representation (32bit/64bit), Error detection and correction.
Unit-2 Register Transfer and Micro-operations: Addition, Subtraction, Multiplication and division algorithms and hardware, Three-state buffer, Binary Adder, Binary Incrementor, Register-transfer language and operations, Arithmetic micro-operations, Logic micro-operations, Shift micro-operations, Arithmetic logic shift unit.
Unit-3 Computer Organization and Design: Instruction codes - Direct and Indirect Address, Computer registers, Computer instructions, Timing and control, Instruction cycle, Instruction format, Memory, register, and input-output reference instructions, Input/ Output and interrupts, Design and working of a complete basic computer, Control functions, Design of accumulator logic.
Unit-4 Memory Organization: Memory hierarchy, High-speed memories, Main Memory, Cache memory, Associative memory, Memory management techniques.

ASSIGNMENT QUESTIONS

Q1. Explain the following:

a) Given the codewords:
C1 = 1101010, C2 = 1110010, C3 = 1001110, C4 = 0101011 and a received word R = 1111110.
Identify the transmitted codeword using Hamming distance.

b) If the multiplicand is **+13 (01101)** and the multiplier is **-6 (11010)**, use Booth's algorithm to compute the product. Show each step (including A, Q, Qn+1, and shift operations).

c) Differentiate addition and subtraction algorithms along with highlighting the important features of both.

Q2. Explain the following:

a) Design a complete basic computer along with its working in detail. Make use of suitable examples.

b) Make an accumulator logic design and explain how it operates. Make use of suitable examples.

c) Analyze how different memory management techniques (like paging and segmentation). influence system performance and memory utilization.